LearnSync AI: AI Personalized Learning Companion for Grade 10 Students

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

The rapid growth of digital learning has made independent study an important part of modern education. Students increasingly use digital learning applications such as review tools, flashcards, and quiz platforms to reinforce lessons outside the classroom. However, many existing educational applications continue to provide standardized, one-size-fits-all learning experiences, presenting similar learning activities regardless of students' individual learning progress, strengths, or areas of difficulty (Bernacki et al., 2021; Raj & Renumol, 2022).

As a result, students may spend unnecessary time reviewing concepts they have already mastered while receiving insufficient support for topics they find challenging. In addition, learning activities that do not appropriately match students' knowledge and skill levels may reduce engagement and contribute to boredom during independent study (Krannich et al., 2025). These challenges highlight the need for more personalized learning tools that can better respond to students' individual learning needs.

The limitations of one-size-fits-all learning tools can be explained through Cognitive Load Theory, which states that students learn more effectively when instructional materials are designed to match their cognitive capacity. Digital learning platforms should therefore be designed to meet students' individual

learning needs, as excessive, irrelevant, or poorly organized information can overload working memory and reduce learning effectiveness (Mayer, 2020)

Well-designed instructional materials minimize unnecessary cognitive load, allowing students to focus on understanding and retaining new knowledge (Castro-Alonso et al., 2021). Likewise, digital learning environments that adapt to students' learning needs can provide more effective and engaging learning experiences by reducing extraneous cognitive load (Skulmowski & Xu, 2022). These findings emphasize the importance of designing educational technologies that personalize learning experiences according to students' current knowledge and learning progress.

Digital learning technologies have significantly improved students' opportunities for independent learning by providing flexible access to educational resources anytime and anywhere. Modern educational applications offer various features such as digital quizzes, flashcards, note-taking tools, and learning analytics to support self-directed learning and encourage students to study beyond the classroom (Viberg et al., 2020).

However, many existing educational applications are designed to support only specific learning activities rather than providing a comprehensive learning environment. As a result, students often need to use multiple applications to meet different learning needs, which can interrupt their learning process and increase unnecessary mental effort. Research emphasizes that integrating learning activities and analytics within a unified digital environment provides a

more effective learning experience and better supports students' self-regulated learning (Pishtari et al., 2020). These findings suggest that while educational technology has expanded opportunities for independent learning, there remains a need for more integrated and personalized learning platforms.

Artificial Intelligence has emerged as a promising technology for overcoming the limitations of conventional digital learning platforms by enabling personalized and adaptive learning experiences. Unlike traditional educational applications that rely on fixed learning activities, AI-powered systems can analyze students' learning progress, identify their strengths and weaknesses, and provide learning paths, instructional content, and feedback that are tailored to individual learning needs (Barrera Castro et al., 2024).

Recent research also highlights that AI-driven adaptive learning technologies improve learning experiences by continuously adjusting instructional support according to learners' performance and progress, allowing students to learn at an appropriate pace while receiving personalized guidance (Bayly-Castaneda et al., 2024). These capabilities transform digital learning platforms from static educational tools into intelligent learning environments that provide continuous and personalized support throughout the learning process.

The client of this study is Pasig Christian Academy, a private educational institution in the Philippines that provides quality basic education to prepare students for higher levels of learning. Among its learners, Grade 10 students face the challenge of preparing for the transition to Senior High School, where lessons

become more specialized and academically demanding. While classroom instruction provides students with the necessary knowledge and guidance, much of their mastery depends on independent study outside school hours. During this period, students receive limited teacher supervision and often rely on digital learning tools that may not adequately address their individual learning needs. These circumstances highlight the need for a learning platform that can provide personalized academic support beyond the classroom.

To address these challenges, the researchers propose LearnSync AI, an educational platform with Artificial Intelligence integration designed to support the independent learning of Grade 10 students. The platform combines Bayesian Knowledge Tracing (BKT) and Generative Artificial Intelligence (Generative AI) to deliver personalized learning experiences based on each learner's progress and estimated mastery. By continuously monitoring students' performance, the system adapts review materials and provides timely feedback that matches individual learning needs. Through this adaptive approach, LearnSync AI aims to provide a more effective and engaging learning experience than traditional one-size-fits-all review platforms.

LearnSync AI integrates multiple evidence-based learning strategies into a single platform to support independent study. It provides a personalized learning analytics dashboard that displays mastery progress, study streaks, total study hours, digital badges, and performance statistics across different learning methods. Students can engage in AI-assisted learning strategies such as the Feynman Technique and Free Recall, where Generative AI provides feedback

and explanations to strengthen conceptual understanding. The platform also includes Adaptive Flashcards and Adaptive Test Mode, which continuously update students' estimated mastery using Bayesian Knowledge Tracing. To encourage consistent engagement, LearnSync AI features AI-generated educational games, including Crossword, Term Matching, and Timed Trivia, whose difficulty is personalized using the learner's current mastery estimate. In addition, the platform supports peer-to-peer resource sharing, allowing students to share learning materials such as flashcards, uploaded resources, adaptive test histories, and selected AI-generated study outputs within the learning community.

By integrating adaptive learning and personalized feedback into a single platform, LearnSync AI provides students with learning activities that better match their individual learning progress and mastery. Rather than relying on fixed review materials, the platform continuously adapts learning experiences to support more efficient and meaningful independent study.

The proposed platform is expected to benefit both students and the institution by supporting more effective independent learning. Students are provided with a personalized learning environment that encourages consistent study habits and self-regulated learning, while teachers may use the platform as a supplementary tool that complements classroom instruction. For the institution, the implementation of innovative educational technologies demonstrates its commitment to improving students' learning experiences through modern digital solutions.

Overall, LearnSync AI is proposed as an integrated educational platform that addresses the limitations of fragmented and one-size-fits-all learning applications. By combining adaptive learning, learning analytics, AI-assisted study strategies, and collaborative learning resources within a single system, the platform aims to provide Grade 10 students with a more personalized, efficient, and engaging environment for independent learning while supporting Pasig Christian Academy's educational objectives.

REVIEW OF RELATED LITERATURE

This chapter presents a comprehensive synthesis of local and international literature, research studies, and technological frameworks relevant to the development of LearnSync AI.

1. Modern Study Challenges and Diverse Student Learners

Developing an effective digital learning platform requires recognizing that students differ in how they learn, process information, and respond to instruction. Learners vary in their prior knowledge, academic abilities, motivation, learning pace, and study strategies. Because of these differences, a single teaching approach may not effectively support all learners. Research on personalized education suggests that learning environments

become more effective when they consider individual learner characteristics and provide appropriate support rather than relying solely on a one-size-fits-all approach (Tetzlaff et al., 2021).

In addition, studies on adaptive learning emphasize that educational systems that respond to learners' needs and progress can create more engaging and effective learning experiences. (Plass & Pawar, 2020). As digital learning continues to expand, educational platforms are increasingly expected to provide flexible and personalized learning experiences that address the diverse needs of students.

1.1. Fast and Slow Learners

Students differ in the speed at which they process information, understand new concepts, and complete learning activities. These differences influence how learners respond to classroom instruction and digital learning environments. In mixed-ability classrooms, some students quickly master lessons and may become less engaged when instruction progresses too slowly, while others require additional time, practice, and guidance before fully understanding the material. Because learners have different academic needs, using the same teaching pace for everyone

may reduce engagement and limit learning opportunities for both high-performing and struggling students. Research on differentiated instruction suggests that learning environments should provide flexible support and varied learning opportunities to accommodate differences in students' abilities and learning pace (Ziernwald et al., 2022).

Likewise, research on educational differentiation indicates that adapting learning experiences to individual learner needs may improve participation, motivation, and academic achievement in diverse classrooms (Eikeland & Ohna, 2022). These findings highlight the importance of designing learning platforms that can support students at different levels of understanding rather than requiring all learners to progress at the same pace.

1.2. Study Preferences

Students have different preferences in how they manage their learning, with many relying on self-directed study to review lessons at their own pace and monitor their academic progress. Self-directed learning encourages learners to take responsibility for their education by selecting appropriate study materials, setting learning goals, and

evaluating their understanding. Research suggests that flexible digital learning environments can promote learner autonomy and support positive online learning experiences (Zhu et al., 2024).

In addition, students can benefit from sharing notes, reviewers, and other learning resources with their peers. Collaborative note-taking and shared learning materials allow learners to access different explanations, improve the completeness of study resources, and reduce cognitive load, leading to more effective learning (Costley & Fanguy, 2021). Therefore, digital learning platforms should support both independent study and educational resource sharing to accommodate diverse learning preferences and encourage continuous learning.

1.3. Digital Overload

The increasing use of digital technologies in education has created learning environments where students often rely on multiple digital tools and platforms to access learning materials, complete assessments, submit assignments, and communicate with instructors and classmates. Managing different interfaces and navigating

between separate systems can increase the mental effort required to perform learning tasks, leaving fewer cognitive resources available for understanding and retaining new information.

Research on digital learning suggests that unnecessary complexity in online learning environments contributes to extraneous cognitive load, which can reduce learning efficiency and make it more difficult for students to focus on instructional content (Skulmowski & Xu, 2021).

Furthermore, learning analytics research highlights the importance of designing well-organized digital learning environments that support students through efficient access to learning resources and educational activities, reducing unnecessary barriers to learning (Ifenthaler & Yau, 2020). These findings emphasize the value of developing integrated learning platforms that centralize educational resources and activities, providing students with a more organized and efficient learning experience.

1.4. Grade 10 Transition

Grade 10 is a crucial stage because it prepares students for the heavier academic demands of Senior High

School, requiring strong reasoning and cognitive skills (Gorosin, 2025). During this final year of Junior High School, core subjects reach their highest level of difficulty. However, this academic transition is often very difficult for learners. Students must shift heavily toward independent learning, and facing these harder subjects without proper guidance can quickly cause mental exhaustion (Bean & Melzer, 2021; Musyadad et al., 2024).

When students struggle with advanced topics at home, standard study methods and rigid quiz apps fail to provide the necessary support. To survive this transition, students need more than just traditional memorization; they require immediate, targeted help that can identify their specific weaknesses. Using smart, adaptive review tools that automatically adjust to a student's individual learning pace helps reinforce core knowledge, prevents study fatigue, and successfully prepares them for Senior High School (Suntharalingam, 2024).

The reviewed literature demonstrates that students differ significantly in their learning pace, study preferences, and academic readiness. Furthermore, they face compounding challenges, such as

cognitive overload from navigating multiple digital platforms and the intense academic pressure of transitioning through Grade 10. These issues highlight critical gaps such as the urgent need for a unified, personalized learning environment that supports diverse learners without adding unnecessary digital fatigue.

LearnSync AI directly addresses these gaps by consolidating the learning experience into a single platform. By integrating personalized systems, it can dynamically adapt to fast and slow learners alike, preventing mental exhaustion. Coupled with gamified activities and peer resource sharing, the platform also provides a comprehensive, targeted study companion that meets the exact academic and psychological needs of Grade 10 students.

2. Analysis of Independent Study Tools and Research Gaps

Independent study tools have become an important part of modern education by providing students with flexible ways to review lessons and learn outside the classroom. Various digital learning applications offer features such as flashcards, quizzes, note-taking, and progress tracking to support self-directed learning (Viberg et al., 2020). Although these tools have expanded access to learning resources, many educational applications are designed to support specific learning activities rather than providing a

comprehensive learning environment. As a result, students often rely on multiple applications to meet different learning needs.

Research emphasizes that effective digital learning environments should integrate learning activities and minimize unnecessary cognitive demands so that students can focus on learning instead of navigating multiple technologies (Skulmowski & Xu, 2022). These observations reveal research gaps in existing independent study tools, particularly in integrating multiple evidence-based learning methods and providing adaptive support within a single platform.

2.1. Fragmentation of Current Educational Applications

Educational applications have become widely used to support independent learning by providing digital tools such as flashcards, quizzes, note-taking, and progress tracking. However, many educational applications are designed to emphasize specific learning activities rather than providing a fully integrated learning environment. As a result, students often rely on different platforms to support various aspects of their independent study. This fragmented learning experience can interrupt the learning process and increase unnecessary mental effort as students switch between

separate tools. Research on digital learning environments emphasizes that well-integrated learning systems reduce unnecessary cognitive demands and provide more effective learning experiences by allowing learners to focus on learning rather than navigating different technologies (Skulmowski & Xu, 2022).

Likewise, studies on learning design highlight the importance of integrating learning activities and analytics within a unified environment to better support students' learning processes and self-regulated learning (Pishtari et al., 2020). These findings suggest that educational platforms should integrate multiple evidence-based learning methods into a single system to provide a more seamless and effective independent learning experience.

2.2. Limitations of Static Review Systems

Many educational applications continue to rely on static review systems, where students receive the same learning materials and review activities regardless of their individual progress or level of understanding. These systems often present fixed quizzes, flashcards, or review sessions without considering students' strengths, weaknesses, or

mastery of different topics. As a result, learners may spend too much time reviewing concepts they already understand while receiving insufficient practice on topics they find difficult.

Research on personalized learning has shown that adaptive learning environments improve the learning experience by adjusting instructional content and learning activities according to students' individual needs, learning pace, and performance (Bernacki et al., 2021). Likewise, adaptive content recommendation systems provide personalized learning resources based on learner characteristics, helping students receive more appropriate learning support than traditional one-size-fits-all review methods (Raj & Renumol, 2022). These findings suggest that static review systems have limitations in supporting personalized learning and highlight the importance of adaptive educational technologies.

The reviewed literature demonstrates that fragmented educational applications and static review systems present challenges to independent learning by increasing unnecessary cognitive demands and limiting opportunities for personalized instruction. Although existing educational

applications effectively support specific learning activities, few integrate multiple evidence-based learning methods with adaptive learning technologies within a single platform.

Guided by these findings, LearnSync AI is designed to address these research gaps by combining cognitive learning strategies, gamification, learning analytics, and adaptive artificial intelligence into one unified learning environment. Through this integrated approach, the platform aims to provide Grade 10 students with a more personalized, efficient, and engaging experience for independent learning.

3. Learning Analytics and Student Motivation

Learning analytics has become an important tool in modern education by providing students with real-time information about their learning progress. Traditionally, educational data such as grades and assessment results were primarily used by teachers to monitor student performance. However, modern learning platforms now present this information directly to students through learner-facing dashboards, allowing them to track their own progress and identify areas that require further study (Matcha et al., 2020; Valle et al., 2021). By giving students access to their own learning data, these systems promote self-regulated learning,

where learners actively monitor, evaluate, and regulate their learning strategies.

Research has shown that learner-facing dashboards increase students' awareness of their academic progress, support self-regulated learning, and improve motivation by helping learners visualize their achievements and learning goals (Aguilar et al., 2021; Paulsen & Lindsay, 2024). Rather than feeling uncertain about their progress, students can make informed decisions about what to review and how to improve their performance. Collectively, these studies highlight the value of integrating learner-facing analytics into digital learning platforms to support students' motivation, self-regulated learning, and academic progress.

3.1. Visualizing Mastery Progress

Visualizing mastery progress helps students monitor their learning by showing their current level of understanding and progress toward learning goals. Unlike traditional grading systems, which often provide feedback after assessments, learner-facing dashboards allow students to monitor their progress continuously. This helps them identify their strengths and the topics that require further study while

developing metacognitive skills such as monitoring and reflecting on their own learning (Matcha et al., 2020).

Research has shown that learner-facing dashboards help students monitor their progress, evaluate their learning strategies, and set learning goals (Valle et al., 2021). IIn addition, providing students with access to their learning data increases their awareness of their academic performance and encourages them to take greater responsibility for their own learning (Aguilar et al., 2021). Collectively, these findings suggest that visualizing mastery progress is an effective way to support independent learning, strengthen students' metacognitive skills, and improve engagement with their studies.

3.2. Study Consistency and Behavioral Engagement

Study streaks and digital badges are features that encourage students to develop consistent learning habits by recognizing their regular participation and achievements. A study streak records the number of consecutive days a student completes learning activities, while digital badges acknowledge milestones such as completing lessons or reaching learning goals.These features provide continuous

feedback and recognition that can motivate students to remain engaged and participate consistently in learning activities (Sailer & Homner, 2020).

Research has shown that consistent engagement and self-monitoring help students develop self-regulated learning skills and improve their persistence in completing learning activities (Matcha et al., 2020). Learning analytics studies also indicate that monitoring students' behavioral engagement helps promote active participation and supports improved learning performance by encouraging consistent involvement in learning activities (Johar et al., 2023).

In addition, achievement indicators such as digital badges have been found to increase learner engagement and encourage continued participation when they are linked to meaningful learning goals and progress (Tahir et al., 2022). Collectively, these findings suggest that study streaks and digital badges can promote consistent learning habits, strengthen learner engagement, and support regular participation in independent learning.

3.3. Time-on-Task and Self-Regulated Learning

Time-on-task refers to the amount of time students spend completing learning activities. In digital learning environments, tracking and displaying study time helps students monitor their learning habits and become more aware of how they manage their time. Unlike traditional learning environments, where students' independent study time is often difficult to monitor, digital platforms provide real-time information that allows learners to evaluate their study pace and identify whether they are spending enough time on their lessons (Hilliger et al., 2021).

Research has shown that monitoring learning behaviors, including time spent on learning activities, supports self-regulated learning by encouraging students to plan, monitor, and reflect on their learning process (Lim et al., 2021). In addition, learning analytics that visualize study behaviors help students become more aware of their learning progress and encourage them to take greater responsibility for their academic performance (Aguilar et al., 2021). These findings suggest that displaying time-on-task information can promote greater responsibility for learning, support independent learning, and encourage students to develop effective study habits.

The reviewed literature demonstrates that providing students with direct access to their own learning data significantly improves their motivation and self-regulated learning. Features such as visualizing mastery progress, tracking study time, and using gamified rewards like streaks and badges help learners build consistent study habits and monitor their own performance. These tools are especially important for students who need to manage their time and identify their own weaknesses during unsupervised study.

Guided by these findings, LearnSync AI features a personalized learning analytics dashboard designed to support Grade 10 students during their independent studies. The platform centralizes their educational data by displaying real-time mastery progress, total study hours, study streaks, and digital badges. By transforming students' learning activities into clear, visual feedback, the platform encourages them to take greater responsibility for their own learning, maintain consistent study routines, and actively monitor their academic progress without requiring continuous teacher supervision.

4. Using Artificial Intelligence for Personalized Student Support

Artificial Intelligence has become an important component of modern educational systems by providing personalized support that addresses the diverse learning needs of students. Traditional

learning approaches often rely on fixed instructional materials that do not adapt to individual differences in learning pace, prior knowledge, or performance. In contrast, AI-based educational systems can analyze student performance and learning behaviors to personalize instructional content, provide timely feedback, and recommend learning activities based on individual progress (Chen et al., 2020). By adapting learning experiences to each student's needs, AI has the potential to improve student engagement, promote deeper understanding, and support more effective learning outcomes.

Beyond personalizing learning content, AI can also provide continuous academic support by offering personalized feedback, recommending appropriate learning activities, and identifying areas where students need additional practice or guidance. Unlike traditional educational platforms that present the same content to all learners, AI-powered systems can respond to individual learning needs by adjusting learning support based on students' progress and performance. This personalized approach enables students to learn at their own pace while receiving timely assistance that strengthens their understanding and encourages continuous learning (Chen et al., 2020).

4.1. AI in Modern Classrooms

The integration of Artificial Intelligence in modern classrooms has played an important role in shifting traditional educational approaches toward more flexible and adaptive learning environments. Historically, many learning systems have followed a one-size-fits-all approach in which students receive the same lessons and review materials regardless of their individual abilities, learning pace, or level of understanding. In contrast, AI-based educational systems can analyze student interactions and performance data to personalize instructional content according to individual learning needs (Chen et al., 2020).

These systems provide personalized feedback, adjust task difficulty, and recommend learning activities based on each student's current level of understanding, allowing learners to receive appropriate support throughout the learning process (Murtaza et al., 2022). This adaptive capability enables students to learn at their own pace while improving engagement, learning efficiency, and independent study.

4.2. AI in Finding Student Learning Gaps

Artificial Intelligence has become an important tool for identifying students' learning gaps and providing personalized academic support. Traditional assessment methods often rely on periodic tests and evaluations, which may not immediately identify areas where students are struggling. In contrast, AI-driven adaptive learning systems can continuously analyze students' learning behaviors and performance data to identify concepts that require additional practice or support (Strielkowski et al., 2024).

By monitoring students' progress throughout the learning process, these systems can provide timely feedback and recommend appropriate learning activities based on individual needs. This enables students to recognize their learning gaps, focus on areas that need improvement, and develop a stronger understanding of concepts while studying independently.

The reviewed literature highlights the important role of Artificial Intelligence in enhancing education by providing personalized learning experiences, adapting instructional content based on students' performance, and identifying learning gaps that require additional support. These capabilities address the limitations of traditional learning

approaches, which often provide the same learning experience to all students regardless of their individual needs and progress.

Guided by these findings, LearnSync AI integrates Artificial Intelligence to deliver adaptive and personalized academic support for Grade 10 students. By continuously adjusting learning activities according to students' progress and identifying areas that require reinforcement, the platform promotes more effective independent learning and encourages deeper conceptual understanding. Through this adaptive approach, LearnSync AI aims to provide a responsive learning environment that improves student engagement, strengthens knowledge retention, and supports better academic performance.

5. Combining Gamification and Cognitive Learning Methods

The combination of gamification and cognitive learning methods has emerged as an effective approach for enhancing student engagement and improving learning outcomes in digital education. Gamification increases learners' motivation by incorporating game elements such as challenges, rewards, and interactive activities into the learning process, encouraging active participation and sustained engagement (Sailer & Homner, 2020).

At the same time, cognitive learning methods emphasize strategies that strengthen understanding, memory, and long-term

knowledge retention through meaningful learning experiences. Research has shown that integrating gamification with cognitive training not only increases learner motivation but also supports cognitive performance by making learning activities more engaging and interactive (Vermeir et al., 2020). By combining motivational game mechanics with evidence-based cognitive learning strategies, digital learning platforms can provide enjoyable learning experiences while promoting deeper understanding, independent learning, and long-term retention of knowledge.

5.1. Gamification Procedures

Gamification has become an effective teaching approach that improves student engagement, motivation, and learning by incorporating game elements into educational activities. Unlike traditional study methods that rely mainly on reading or memorization, gamified learning encourages students to participate actively through interactive challenges, immediate feedback, and achievable goals. Studies have shown that using gamification in learning environments can increase students' motivation, engagement, and academic performance (Sailer & Homner, 2020).

In digital learning platforms, activities such as crossword puzzles, timed trivia quizzes, and matching card games make studying more enjoyable while encouraging students to recall information and reinforce what they have learned. By combining game elements with educational content, these activities create a more interactive learning experience that supports independent learning and improves long-term knowledge retention.

5.1.1. Crossword

Crossword puzzles have become a useful educational activity for reinforcing knowledge and improving students' understanding of key concepts. Unlike traditional review methods that rely solely on reading or memorization, crossword puzzles encourage learners to actively recall information by identifying and connecting terms with their corresponding definitions or clues. This process promotes deeper cognitive engagement while making learning more interactive and enjoyable.

Crossword puzzles have a positive impact on learning by improving students' knowledge and

supporting better performance in objective assessments (Arnold et al., 2024). These findings suggest that incorporating crossword puzzles into digital learning platforms can strengthen concept retention while increasing student engagement during independent study.

5.1.2. Matching Card Games

Matching card games are widely used as interactive learning activities that promote active participation and reinforce students' understanding of concepts. By requiring learners to match related terms, definitions, images, or concepts, these activities encourage active recall and strengthen associations between pieces of information rather than relying on passive reading.

Recent research found that matching card-based learning materials significantly improved students' vocabulary mastery and learning outcomes by providing an engaging and interactive learning experience (Febrianti & Shofiyuddin, 2025). Because matching activities encourage repeated retrieval

practice and concept association, their underlying principles can be adapted to reinforce foundational knowledge in digital learning environments.

5.1.3. Timed Trivia Questions

Timed trivia questions are an effective gamified learning activity that combines retrieval practice with interactive quiz-based learning. By requiring learners to answer questions within a limited time, students actively retrieve information from memory while maintaining focus under structured time constraints. This transforms conventional review sessions into engaging learning experiences that encourage continuous participation and immediate feedback. A literature review found that game-based quiz platforms such as Kahoot! consistently improve student engagement, motivation, classroom participation, and learning performance across different educational settings (Wang & Tahir, 2020).

In addition, the integration of game elements such as time limits, scores, and immediate feedback has been shown to positively influence learners'

motivation and academic outcomes when appropriately incorporated into educational activities (Sailer & Homner, 2020). Together, these findings suggest that timed trivia questions provide an engaging review strategy that encourages active participation while reinforcing previously learned concepts through repeated retrieval practice.

5.2. Retrieval Practices

Cognitive retrieval practices are an important component of modern educational psychology that enhance how students recall and retain learned information. Unlike traditional study methods that rely on rereading or passive memorization, retrieval-based learning requires students to actively recall information from memory, which strengthens long-term retention and improves understanding of concepts. Research has consistently shown that retrieval practice is more effective than passive review strategies in improving learning outcomes and academic performance (Agarwal et al., 2021).

In addition, testing-based learning has been found to significantly enhance classroom learning by improving

memory retention and reinforcing conceptual understanding over time (Yang et al., 2021). Techniques such as free recall and structured explanation exercises, including the Feynman Technique, are commonly used applications of retrieval practice that encourage learners to actively reconstruct knowledge from memory. Through repeated retrieval of learned material, students are able to identify areas that require further review and strengthen their overall understanding of lessons during independent study.

5.2.1. Feynman Technique

The Feynman Technique is a learning strategy that helps students understand lessons by explaining concepts in their own words using simple language. Instead of memorizing information, students actively recall what they have learned, identify parts they do not fully understand, and review those topics until they can explain them clearly. This process improves understanding and strengthens long-term memory.

A study found that students who used the Feynman Technique achieved higher posttest scores and greater learning gains than those who used traditional learning

methods (Reyes et al., 2021). The researchers concluded that explaining concepts in simple terms helps students develop a deeper understanding, become more independent learners, and improve their academic performance. These findings suggest that the Feynman Technique is an effective cognitive learning strategy that can improve comprehension and knowledge retention.

5.2.2. Free Recall

Free recall is a form of retrieval practice that helps students strengthen their memory by encouraging them to remember and explain what they have learned without using notes or other learning materials. Instead of simply rereading information, students actively retrieve knowledge from memory, making it easier to identify topics they understand and concepts that require further review. This process improves understanding and supports long-term retention (Yang et al., 2021).

Research has shown that retrieval practice, including free recall, is more effective than passive study methods such as rereading for improving learning and memory (Agarwal et al., 2021; Yang et al., 2021). Actively recalling

information improves memory, strengthens understanding, and leads to better academic performance (Agarwal et al., 2021). Likewise, retrieval-based learning helps students retain information longer and reinforces their understanding of lesson concepts (Yang et al., 2021). These findings suggest that free recall is an effective cognitive learning strategy for improving students' learning outcomes.

5.3. Adaptive Flashcards

Flashcards are widely used as a learning tool that supports active recall and improves memory retention. Unlike passive study methods such as rereading notes, flashcards require learners to actively retrieve information from memory, which strengthens understanding and reinforces learned concepts. Research on vocabulary learning has shown that intentional use of flashcards can significantly improve recall performance and learning outcomes compared to traditional study methods such as word lists and repeated reading (Webb et al., 2020).

In addition, studies on digital and mobile-assisted learning indicate that flashcards are effective in supporting vocabulary acquisition and improving students' retention of

academic words when used as part of structured study activities (Xodabande et al., 2022). The effectiveness of flashcards can be further enhanced when combined with spaced repetition principles, where review sessions are scheduled at increasing intervals based on learner performance.

Modern adaptive learning systems further improve this process by using data-driven approaches to personalize review schedules. Unlike static study methods where learners may repeatedly review familiar material or neglect difficult concepts, adaptive flashcard systems adjust the frequency of review based on student performance. These systems help optimize learning efficiency by focusing more on challenging items and reducing unnecessary repetition, which can improve long-term retention and reduce cognitive load (Huang, 2025).

5.4. Adaptive Assessment

Computerized adaptive testing has become an important development in modern educational assessment, shifting traditional fixed testing formats toward more flexible and data-driven evaluation systems. Unlike conventional assessments where all students answer the same set of

questions regardless of ability, adaptive testing systems adjust question difficulty based on learners' responses and performance patterns. Recent research highlights that these systems use real-time performance data and machine learning techniques to select appropriate test items, allowing assessments to better match each student's level of understanding (Zhuang et al., 2024).

In addition, AI-driven quiz generation systems have been shown to support the creation of dynamic assessments that can adjust question difficulty and provide more personalized evaluation experiences (Al-Gburi, 2025). Through these adaptive mechanisms, computerized testing systems aim to improve the efficiency of assessments and provide more accurate representations of student knowledge and learning progress.

The reviewed literature suggests that integrating gamification, retrieval-based learning strategies, flashcards, and adaptive assessment can contribute to more effective and engaging learning experiences compared to traditional passive study methods. Gamification has been shown to enhance student motivation and engagement by incorporating interactive and goal-oriented learning activities, while cognitive strategies

such as active recall and retrieval practice support deeper understanding and long-term retention of information. Similarly, adaptive flashcards reinforce learning through repeated exposure and retrieval of information, which helps strengthen memory and improve recall over time. In addition, adaptive assessment approaches improve learning efficiency by adjusting the difficulty and content of evaluation tasks based on student performance, allowing learners to be assessed according to their current level of understanding.

Collectively, these approaches highlight the value of combining motivational and cognitive learning strategies within digital learning environments. In relation to these findings, LearnSync AI is designed to integrate these principles into a unified system that supports personalized, adaptive, and self-directed learning, with the aim of improving student engagement, knowledge retention, and academic performance.

6. Performance Evaluation of Language Models and Inference Systems

The performance of large language models and their inference systems is an important factor in ensuring that AI-powered educational platforms provide fast, reliable, and responsive learning experiences. In digital learning environments, delays in processing user inputs or generating responses can

negatively affect student engagement and disrupt learning continuity. As a result, modern LLM systems are designed with optimized inference and serving architectures that improve how models handle multiple user requests efficiently. Research on distributed model serving systems shows that optimized inference techniques can improve system throughput and response efficiency in transformer-based language models (Yu et al., 2022).

In addition, studies on memory-efficient LLM serving highlight that improved resource management and computational optimization help reduce latency and support smoother real-time interactions in large-scale AI applications (Kwon et al., 2023). These advancements are particularly important for educational platforms that rely on generative AI, as they ensure timely feedback, stable performance, and uninterrupted learning support for users.

6.1. LLM in Materials and Context

The use of large language models (LLMs) in educational systems plays an important role in improving how students interact with complex and lengthy learning materials. Early transformer-based language models demonstrated strong capabilities in understanding and

generating human-like text, enabling flexible language processing across a wide range of inputs (Brown et al., 2020).

While earlier transformer-based language models were limited by relatively restrictive context windows that affected continuity in multi-topic interactions, recent advancements in large language models have significantly improved their ability to process longer contextual inputs. These improvements allow models to generate more coherent and context-aware responses when working with extended learning materials. As a result, when integrated into educational platforms, LLM-based systems can better support student understanding of complex topics by providing more consistent and contextually relevant explanations during independent study (Sharma et al., 2025).

6.2. Real Time Feedback

Large language model–based systems play an important role in providing immediate formative feedback within digital learning environments. In traditional systems, delays in generating responses may interrupt students' focus

during active learning tasks such as retrieval practice and review activities. In contrast, modern AI-powered systems are capable of delivering near real-time feedback and personalized guidance, which supports student engagement and sustained participation in learning activities (Kasneci et al., 2023).

Reducing response delays also helps create a smoother interaction between the learner and the system, minimizing disruption during study sessions. This immediate feedback allows students to quickly identify and correct mistakes, which can improve learning efficiency, task engagement, and revision behavior (Meyer et al., 2024).

6.3. Model Specialization and Dynamic Routing

The use of multiple large language models within a single system plays an important role in improving the efficiency and responsiveness of AI-powered applications. Instead of relying on a single model to handle all types of tasks, modern systems can distribute requests by assigning them to different models based on task complexity and computational requirements. Research on LLM routing systems shows that queries can be directed to different

models depending on expected performance needs, allowing systems to balance quality and efficiency (Ong et al., 2024).

In this approach, simpler or less computationally demanding tasks can be handled by more efficient models, while more complex requests are processed by higher-capacity models. This routing strategy helps optimize system performance by reducing unnecessary computational load and improving response time, resulting in a smoother and more efficient user experience in real-time applications.

The reviewed studies demonstrate that an effective AI-powered educational platform depends not only on the capabilities of large language models but also on efficient inference systems, real-time feedback, contextual understanding, and appropriate task distribution among AI components. Together, these technologies create a learning environment that is responsive, reliable, and capable of supporting students throughout their independent study sessions.

Guided by these findings, LearnSync AI integrates these concepts to provide fast AI-generated assistance, process lengthy study materials effectively, deliver immediate personalized feedback, and maintain smooth system performance. By combining this into a unified application, it provides Grade 10 students with a more efficient, engaging, and

personalized learning experience that supports independent study and improves overall learning outcomes.

7. Bayesian Knowledge Tracing

Bayesian Knowledge Tracing (BKT) is a probabilistic model used in intelligent tutoring systems to estimate a learner's mastery of specific skills based on their sequence of correct and incorrect responses. It works by continuously updating the probability that a student has learned a concept, making it useful for modeling knowledge progression in adaptive learning environments. BKT is commonly used in educational systems due to its simplicity, interpretability, and ability to support dynamic learner modeling without requiring high computational complexity (Bulut et al., 2023).

Recent implementations such as pyBKT further demonstrate its practicality by providing accessible tools for applying BKT models in educational research and learning systems (Badrinath et al., 2021). Although more complex models exist, BKT remains widely used in many educational contexts because it offers an effective balance between computational efficiency and meaningful learner tracking, making it suitable for adaptive learning applications.

7.1. Tracking Models: BKT, DKT, and PFA

In educational data mining, predicting student learning progression involves modeling hidden knowledge states using different algorithmic approaches, including Bayesian Knowledge Tracing (BKT), Performance Factors Analysis (PFA), and Deep Knowledge Tracing (DKT) (Abdelrahman et al., 2022). These models vary in structure and complexity. BKT is a probabilistic model that estimates student mastery based on sequential performance, while PFA is a statistical approach that predicts learning outcomes by considering the cumulative count of prior successes and failures in skill practice. In contrast, DKT utilizes deep learning techniques to model complex learning patterns from large-scale datasets.

Research indicates that deep learning-based approaches such as DKT can achieve strong predictive performance, particularly when large datasets are available, but they may introduce challenges related to interpretability compared to traditional models (Scruggs et al., 2020). Meanwhile, BKT and PFA remain widely used in educational systems due to their simplicity, interpretability, and lower computational requirements, making them suitable for applications that require transparent learner modeling. These differences highlight the trade-offs between predictive accuracy, interpretability, and computational complexity when selecting knowledge tracing models for educational applications

7.2. BKT Selection

Among the different knowledge tracing models reviewed, including Performance Factors Analysis (PFA) and Deep Knowledge Tracing (DKT), the researchers selected Bayesian Knowledge Tracing (BKT) as the learner modeling approach for the system. This decision is based on its suitability for adaptive learning environments that require efficient, interpretable, and sequential tracking of student knowledge.

Unlike DKT, which requires large datasets and higher computational resources, and PFA, which focuses on aggregated performance counts rather than step-by-step learning progression, BKT provides a lightweight probabilistic framework that updates learner mastery based on individual responses. Its simplicity and transparency make it appropriate for educational applications where clear interpretation of student progress is important. Therefore, BKT was chosen to support the system's need for continuous, real-time, and interpretable tracking of learner performance in adaptive learning activities such as flashcards and assessments.

7.3. BKT Parameters

The Bayesian Knowledge Tracing (BKT) framework operates using four key and interpretable parameters that are used to estimate a learner's knowledge state after each interaction (Badrinath et al., 2021; Bulut et al., 2023). These include two learning-related parameters namely *prior knowledge,* which represents the probability that a student has already mastered a skill before interacting with the system, and the *learning rate,* which indicates the probability that a student acquires the skill after a learning opportunity.

In addition, BKT incorporates two performance-related parameters namely the *guess probability*, which accounts for the likelihood that a student answers correctly despite not having mastered the concept, and the *slip probability*, which captures the chance of an incorrect response despite the student having already learned the skill. These parameters allow BKT to more accurately model student learning behavior by distinguishing true understanding from random or careless responses.

The integration of BKT into LearnSync AI provides a highly efficient method for tracking student mastery during independent study sessions.

By continuously updating a learner's knowledge state based on their real-time responses, the system can accurately estimate skill mastery while accounting for random errors such as guessing or slipping. Relying on this lightweight tracking infrastructure ensures that the application remains fast and responsive.

Ultimately, leveraging BKT enables LearnSync AI to effectively personalize learning activities, dynamically adjust difficulty levels, and provide immediate adaptive feedback, resulting in a highly individualized educational experience.

SIGNIFICANCE OF THE STUDY

This study is significant because it proposes LearnSync AI to provide personalized and adaptive learning for Grade 10 students. The proposed system is expected to benefit the following:

Pasig Christian Academy. The proposed system will benefit Pasig Christian Academy by providing a centralized learning platform that supports students' independent study. Through LearnSync AI, the school can offer AI-assisted study methods, learning progress tracking, and peer-to-peer resource sharing in one platform. This can help improve students' study habits, support

technology-enhanced learning, and strengthen the school's reputation as a modern educational institution.

Educators. The proposed system will benefit educators by providing a supplementary learning tool that students can use outside the classroom. Through personalized study activities and AI-assisted learning, students can strengthen their understanding of lessons independently. This allows educators to focus more on reinforcing important concepts during class while encouraging students to develop better study habits.

Students. The proposed system provides a smart, personalized study companion that adjusts to how fast a student learns. Instead of using generic reviewers, students get adaptive flashcards, AI-guided feedback, and gamified reviews that prevent boredom and confusion. It also allows them to easily share study materials with their classmates to help each other review.

Grade 10 Students. The proposed system specifically targets Grade 10 students because they face the highest level of subject complexity in Junior High School while simultaneously preparing for the critical transition to Senior High School. The platform directly reduces this academic pressure by acting as an AI Personalized at-home tutor, helping them master advanced topics efficiently.

Researchers. This study will help the researchers improve their knowledge and skills in software development, educational technology, artificial intelligence, adaptive learning systems, and Bayesian Knowledge Tracing. It will also provide valuable experience in designing, developing, and implementing an AI-powered educational platform.

Future Researchers. This study may serve as a reference for future researchers who wish to develop AI-powered educational systems. It can also be used as a foundation for improving adaptive learning features, integrating new AI technologies, and expanding the platform for different grade levels and educational settings.

STATEMENT OF THE PROBLEM

While Pasig Christian Academy aims to improve education for Grade 10 students, many still struggle with generic, disconnected study apps that make independent learning ineffective. Research shows that AI can solve this by tailoring education to individual needs.(Kasneci et al., 2023; Tlili et al., 2023). To fix this problem, the researchers propose LearnSync AI, an educational platform that combines Generative AI and Bayesian Knowledge Tracing to provide a truly personalized study companion for students.

To evaluate the platform's technical accuracy, educational effectiveness, and user acceptance, this study will address the following specific research questions:

1. What are the existing challenges encountered by Grade 10 students in traditional learning environments that can be addressed through LearnSync AI?

2. How accurately does LearnSync AI synthesize Generative Artificial Intelligence and Bayesian Knowledge Tracing to optimize its algorithmic outputs?

3. Does LearnSync AI improve Grade 10 students' engagement and independent study habits through AI-assisted learning, adaptive methods, gamification, and peer-to-peer resource sharing?

4. How effective is LearnSync AI in improving the academic performance of Grade 10 students based on pre-test and post-test results?

5. What is the level of acceptability of LearnSync AI based on the ISO/IEC 25010 software quality characteristics as evaluated by Grade 10 students?

SCOPE AND LIMITATIONS

This section defines the precise boundaries of the study. The research focuses on the design, development, and user testing of LearnSync AI to measure the effectiveness of AI-driven personalization, gamification, and collaboration in an educational setting.

Scope of the Study:

This study focuses on the engineering, development, and validation of LearnSync AI, a responsive web application evaluated under ISO/IEC 25010 standards for Functional Suitability, Usability, Performance Efficiency, and Security. System Implementation and Evaluation is strictly limited to 18 Grade 10 students at Pasig Christian Academy for the academic year 2026-2027, chosen for their cognitive maturity and their need to prepare for the rigorous academic complexities of Senior High School. The platform allows students to upload standard secondary education modules as plain text PDFs (maximum 20 MB) into a digital repository.

Central to the system is the classical four-parameter Bayesian Knowledge Tracing (BKT) algorithm, which calculates real-time mastery scores based on student performance in Adaptive Flashcards and Tests. These mastery scores dynamically calibrate the difficulty of AI-generated games (such as Crosswords, Term Matching, and Timed Trivia) and adjust the feedback complexity within the LLM-evaluated Feynman and Free Recall modes. Finally, the application includes a peer-to-peer collaboration layer, enabling students to easily share custom flashcard sets, read-only session histories, and uploaded learning resources with their classmates.

Limitations of the Study:

To maintain the project's feasibility and focus, several limitations are established. System Implementation and Evaluation is restricted to 2 - 4 weeks, which excludes the measurement of long-term knowledge retention or multi-year behavioral changes. Technically, the system faces an algorithmic "cold-start" upon new module uploads, requiring the generation of baseline questions to establish an initial mastery level before adaptation can occur. The AI is optimized exclusively for standard Junior High School core

lessons, restricting the use of tertiary or hyper-specialized professional datasets.

Furthermore, the system strictly utilizes classical BKT, and its mastery scores act entirely independently of the school's official report card grades. To preserve the integrity of academic tracking, student performance within gamified modules does not update the underlying BKT scores, preventing lucky guesses from corrupting the data. Operationally, the application requires a constant internet connection and only accepts plain text PDFs, excluding offline capabilities, voice interactions, and handwritten text recognition.

Built strictly for educational evaluation, the platform excludes commercial components such as payment gateways, automated billing, and enterprise server auto-scaling. Finally, the system strictly complies with the Data Privacy Act of 2012 (RA 10173) by solely collecting the basic academic data necessary for AI generation and tracking, ensuring no highly sensitive personal details are gathered and that student information is never shared or sold to third parties.

CHAPTER II

INTRODUCTION

This chapter presents the methodology used to design, develop, and evaluate LearnSync AI: AI Personalized Learning Companion for Grade 10 Students. This web-based platform is created exclusively for Grade 10 students to support and improve their self-paced, independent study habits. The sections in this chapter detail the system's technical design, development workflows, algorithms, and data structures that allow the platform to operate safely, securely, and efficiently.

To keep the platform focused entirely on the learner, the application is designed with a single user role: the student. Students have full control over their learning experience. They upload their own reference materials to set up private, adaptive study spaces. As they review, the system tracks their performance using Bayesian Knowledge Tracing (BKT) to automatically tailor practice questions to their individual learning needs. Additionally, the platform incorporates the gamification module to sustain student motivation and a secure peer-to-peer workspace sharing utility that allows classmates to easily share and study together.

Finally, this chapter outlines the validation process designed to measure the application’s real-world performance. A two-to-four-week System

Implementation and Evaluation will be conducted at Pasig Christian Academy, involving 18 Grade 10 students as the sole participants. The feedback gathered from these students will be quantified using a software quality evaluation survey based on the ISO/IEC 25010 framework. . This evaluation will formally assess the system's functional suitability, usability, performance efficiency, and security to determine its overall effectiveness and readiness for student use.

RESEARCH DESIGN

This study uses a developmental research design with a quantitative approach to guide the design, construction, and evaluation of the LearnSync AI web application. In educational technology, developmental research focuses on the systematic creation, testing, and evaluation of instructional tools to ensure they are stable, functional, and effective (Angeles & Cortez, 2023). By using this design, this project transitions from theoretical software planning to a functional, real-world application tested by its target users. The systematic recording of the design and development processes allows the research team to translate educational needs into functional software features while collecting objective numbers to prove the application's overall performance.

The developmental process for LearnSync AI is structured into three distinct, student-centered phases. The first phase is the analysis and design phase, which focuses on identifying the specific study challenges faced by Grade 10 students during self-paced study. The findings from this initial phase are used

to design the user interface layout of the Student Portal, map out the architecture of the private study spaces, establish the Bayesian Knowledge Tracing routing rules, and draft the mechanics for the "PlayMe" gamification. The second phase is the development phase, which involves the technical implementation of the web application. Key tasks during this period include configuring the Supabase relational database, establishing PostgreSQL Row-Level Security policies to protect individual student data, programming the generative artificial intelligence parser to extract study terms from student uploads, implementing the real-time Bayesian Knowledge Tracing scoring engine, and developing the peer-to-peer sharing utility. The third and final phase of this process is the evaluation phase, which is designed to test the operational quality and educational impact of the finished application. The platform will be deployed and evaluated over a two-to-four week period, involving a sample size of 18 Grade 10 students at Pasig Christian Academy. This target user group will interact with the application independently, allowing the system to gather operational data and compile user feedback in a realistic self-study environment.

To objectively measure success without bias, the study relies entirely on quantitative data collection through two distinct methods. The first method is quasi-experimental cognitive testing. To measure academic progress, the research team designs, validates, and administers both the pre-test and post-test directly. Before students use LearnSync AI, the researchers administer a baseline diagnostic pre-test. After the pilot study, the researchers personally administer the corresponding post-test. Comparing these test results

mathematically proves whether the adaptive study features actually improve student performance.

The second quantitative data collection method is a system evaluation based on the ISO/IEC 25010 software quality standard. At the end of the pilot period, the 18 student participants will answer a descriptive software quality survey where their feedback is captured using a 5 point Likert scale. This scale converts user ratings into numerical averages, allowing the research team to formally verify the system's performance. The final statistical data will objectively measure and confirm the web application's stability and execution across four core metrics, namely Functional Suitability, Usability, Performance Efficiency, and Security.

ARCHITECTURAL FRAMEWORK

The architectural framework of LearnSync AI is structured as a student-centric, three-tier model designed to provide a secure, personal, and highly adaptive self-study environment. The system isolates individual learning activities, processes behavioral tracking parameters, and dynamically scales challenge levels. The system is split into three main layers: the Presentation Layer, the Application Layer, and the Data Tier.

1. **Presentation Layer -** This layer represents the client-facing user interface running natively inside the student's web browser. It serves as the primary gateway for data ingestion while rendering processed data back to the learner. Within this tier, the Study Space Hub handles the direct capture of the student's raw data inputs. This includes the upload of raw PDF documents (up to 20 MB), qualitative memory retrieval ratings recorded within the flashcard mode (where students rate cards as Easy, Medium, or Hard), unstructured textual inputs for Feynman Technique and Free Recall exercises, and student-configured practice test parameters alongside submitted answers (where students can select one or more test types like multiple choice, identification, matching type, fill in the blank, and true or false). The PlayMe Portal intercepts all user gameplay telemetry, tracking interaction timings and activity logs during Crosswords, Term Matching, and Timed Trivia modules. Finally, the Dashboard Screen manages the visual output stream of the layer, rendering real-time performance analytics, cumulative study hours, active daily streaks, mastery curves, and unlocked milestone badges directly onto the interface.

2. **Application Layer -** This layer operates as the central computational brain of LearnSync AI, intercepting the raw data parameters flowing from the user interface and processing them through specialized algorithmic logic cores. The first major component, the Formative Content & AI Evaluation Core, ingests raw student materials via the Formative Parsing

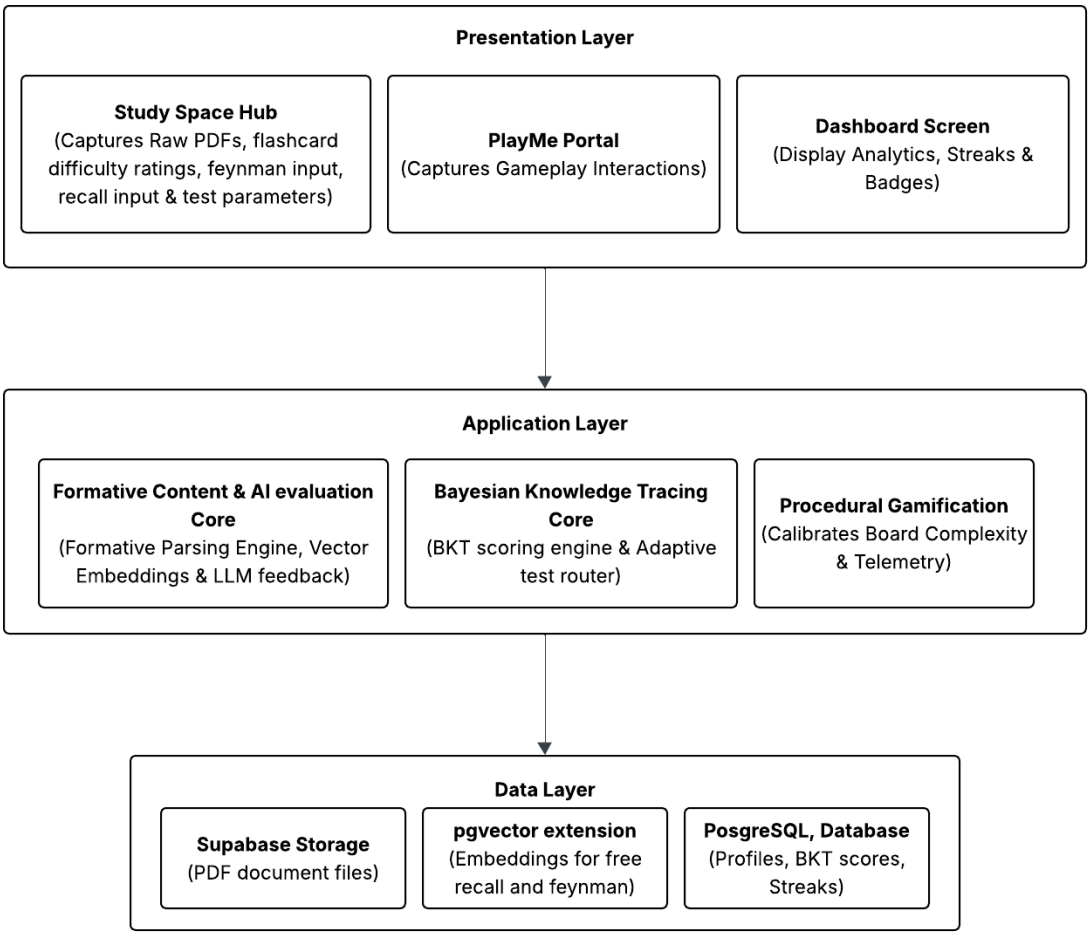
Engine to extract glossaries and core concepts, while utilizing Vector Embeddings and large language model (LLM) feedback to analyze Free Recall and Feynman summaries.

Positioned alongside this is the Bayesian Knowledge Tracing (BKT) Core, which acts as the system's primary cognitive tracking engine. This core processes the ongoing qualitative ratings from the flashcard review sessions and practice test submissions, converting them into standard binary inputs to mathematically update the student's latent mastery score. The Adaptive Test Router then reads this updated mastery state to instantly scale the difficulty of generated practice test questions up or down. To tie the logic layer together, an algorithmic link feeds this live mastery baseline directly into the Procedural Gamification Engine, enabling it to read the calculated knowledge state as a configuration metric to calibrate board complexity and match game difficulty perfectly to the student's active level of understanding.

3. **Data Tier -** The Data layer coordinates all persistent data operations, storage management, and structural security boundaries across the cloud infrastructure. Powered entirely by a unified Supabase platform, this tier secures and isolates information across three specialized storage vectors. The Supabase Storage engine operates as a secure file-bucket repository that houses raw PDF document files uploaded by students for their respective study spaces. Working in parallel is the pgvector extension, a

high-dimensional database module that stores semantic embeddings derived from the student's learning materials to execute fast vector similarity queries for Free Recall evaluations. Lastly, the PostgreSQL Database serves as the structured relational foundation of the entire system, maintaining long-term records of student profiles, updated BKT mastery percentages, and study streak metadata, all while enforcing strict Row-Level Security (RLS) policies to prevent unauthorized data access across individual user profiles.

Figure 1. *Architectural Framework*



ALGORITHM

This section details the mathematical and algorithmic foundation of LearnSync AI's cognitive modeling core. To deliver a highly personalized, self-directed learning experience, the platform integrates the classical four-parameter Bayesian Knowledge Tracing (BKT) model. Originally formalized by Corbett and Anderson (1994) and optimized for modern web architectures using the open-source pyBKT standard (Bulut et al., 2023), BKT operates as a dynamic, hidden Markov model (HMM) with observable nodes. Within LearnSync AI, this algorithm continuously analyzes a student's sequential interactions to

trace their latent knowledge state for any given concept without requiring computationally heavy server resources.

1. **BKT Mathematical Model** - The primary objective of the BKT algorithm is to estimate the hidden probability that a student has mastered a specific concept based on their chronological sequence of correct and incorrect answers. The mathematical framework is governed by four distinct, highly interpretable parameters categorized into two knowledge parameters and two performance parameters:

    a. **Initial Knowledge State** $P(L_0)$: The prior probability that a student already knows or has mastered a concept before interacting with any formative study modes or assessments on the platform.

    b. **Transition Probability** $P(T)$: The learning rate, which represents the probability that a student transitions from an unmastered (not-known) state to a mastered (known) state after experiencing a learning opportunity.

    c. **Guess Probability** $P(G)$: The probability that a student who has not mastered the concept still manages to answer a practice question or flip card correctly.

    d. **Slip Probability** $P(S)$: The probability that a student who has mastered the concept still answers incorrectly due to carelessness, cognitive fatigue, or a reading error.

As the student interacts with the application, their performance at any given learning opportunity *t* is captured as an observable binary variable (obs_t), where an answer is marked as correct ($obs_t = 1$) or incorrect ($obs_t = 0$). When a new observation is captured, the system executes a two-phase Bayesian update. First, the algorithm applies the evidence update phase to calculate the conditional probability of mastery ($P(L_t \mid obs_t)$ based on the observation:

Formula when the students attempt is correct ($obs_t = 1$) :

$$P(L_t \mid obs_t = 1) = \frac{P(L_t)\,(1-P(S))}{P(L_t)\,(1-P(S)) + (1-P(L_t))\,P(G)}$$

Formula when the students attempt is incorrect ($obs_t = 0$) :

$$P(L_t \mid obs_t = 0) = \frac{P(L_t)\,P(S)}{P(L_t)\,P(S) + (1-P(L_t))\,(1-P(G))}$$

Second, the algorithm performs the transition update phase. This step calculates the projected probability of mastery for the subsequent learning opportunity ($P(L_{t+1})$) by factoring in the learning rate ($P(T)$) as the student moves to the next card or question, assuming immediate formative feedback is processed:

$$P(L_{t+1}) = P(L_t \mid obs_t) + (1 - P(L_t \mid obs_t))\,P(T)$$

2. **Algorithmic Implementation in Flashcard Mode -** Within the study space, the spaced repetition manages how the flashcards behaves while

bkt calculates the students mastery of a concept. These flashcards can be made by the student or generated by the AI model using the uploaded files. When a student opens a flashcard session, the system loads their current progress from the database. If it is a brand-new card set, the system initializes their score to the standard starting level $P(L_0)$. During study, the student flips a card to reveal its definition and rates how well they remembered it by choosing Easy, Medium, or Hard. Choosing Easy means the student remembered the card easily. The system schedules this card to reappear in four days and records a correct answer $(obs_t = 1)$. To reflect this high confidence, the algorithm drops the guess rate to a very low number $(P(G_{easy}) = 0.05)$. This tells the formula that luck was not involved, giving the student a large boost in their mastery score. Choosing medium means the student remembered the card but had to think about it. The system schedules this card to reappear in three days and also records as correct answer $(obs_t = 1)$. However, to account for the chance that the student got it right by guessing, the algorithm uses a higher guess rate $(P(G_{medium}) = 0.25)$, which gives them a smaller and more cautious boost to their mastery score. Finally, choosing Hard means the student forgot the card. The system schedules the card to reappear in one day for fast practice and records an incorrect answer $obs_t = 0$. The application sends these ratings directly to the BKT math engine to calculate the student's new progress percentage $P(L_{t+1})$

and saves it in the database. This hybrid setup gives students personalized review times while the BKT engine accurately updates their knowledge statistics behind the scenes.

3. **Algorithmic Implementation in Adaptive test Mode** - Within the adaptive test mode, the Bayesian Knowledge Tracing model serves as the main engine for calibrating the difficulty of generated assessment sessions. When entering Test Mode, the student selects one or more custom question formats, such as multiple choice, identification, matching type, fill in the blank, or true or false. As the student completes their attempts during active testing, their performance is graded as a binary value (1 for correct or 0 for incorrect), and the BKT update pipeline calculates their dynamic mastery score on a per-question basis. This real-time calculated score updates the student's concept-specific history in the central database.

   When the student subsequently requests the generation of a new practice test session on the same topic, the system queries this historical BKT mastery probability to scale the conceptual difficulty of the newly generated items. The adaptive test generation engine utilizes specific mastery thresholds to adjust the generation prompt instructions sent to the large language model.

If the current running mastery probability is high specifically at or above 0.85, the AI model is instructed to generate advanced-tier questions focusing on analytical scenarios, synthesis, and edge-cases within the student's selected formats. If the mastery probability lies in the intermediate range between 0.50 and 0.85, the prompt parameters dictate intermediate-tier questions emphasizing conceptual comparisons, multi-concept recall, and practical applications. Now, if the running mastery sits in the foundational range below 0.50, the generation instructions shift to produce simpler, highly supportive retrieval questions focusing on direct definitions and basic conceptual associations. This session-based adaptation allows students to retain complete control over their preferred test layouts while the system continuously adjusts the cognitive demands of the content to match their developmental needs.

4. **Document Mastery and Gamification Calibration** - While Bayesian Knowledge Tracing tracks individual concepts as isolated, high-precision knowledge components in the database, the system must provide a consolidated progress index for end-users. When a student uploads a study document, the document ingestion engine parses and extracts a set of *N* distinct concept nodes. To render a high-readability overview on the student's dashboard, the application layer dynamically aggregates these independent probabilities using an arithmetic mean. The overall mastery rate M_{doc} for an uploaded document containing *N* concepts is calculated at

any given timestamp using the most current running BKT metrics fetched from the database:

$$M_{doc} = \frac{\Sigma P(L_{current,i})}{N}$$

Where:

- M_{doc} represents the overall document mastery percentage displayed as a progress bar on the user dashboard.
- N represents the total number of distinct concepts extracted from the uploaded document file.
- $P(L_{current,i})$ represents the real-time, running BKT mastery probability of the i-th concept stored under the student's profile in the database. This value is dynamically calculated for each individual concept *i* through the sequential, chronological application of the core BKT equations detailed in *BKT Mathematical Model.* At the initial state (before any student interactions, where *t* = 0), the concept's probability is initialized to the baseline prior $P(L_0)$. For each subsequent practice attempt *t* on that concept, the system captures the binary user interaction response $obs_t = 0$ or $obs_t = 1$, calculates the intermediate conditional probability using the corresponding evidence formula, and executes the transition update to produce the updated probability state $P(L_{t+1})$. The active

value of $P(L_{current,i})$ referenced in this macro aggregation formula represents the most recently computed and saved $P(L_{t+1})$ value stored within the student's concept profile in the database, ensuring that the aggregate overview dynamically adapts to their live learning progress.

This aggregated score is directly utilized by the system's gamification module to calibrate the operational difficulty of generated games (crosswords, term matching, and timed trivia). When a game session is initialized, the system retrieves the current M_{doc} value from the database and applies it as a baseline parameter to the generative AI prompt. For example, if a student's document mastery sits at exactly 50%, the game engine utilizes this value to restrict the vocabulary complexity, clue difficulty, and trivia response timers to align exactly with that 50% knowledge threshold. This integration protects students from cognitive frustration while keeping gamified learning tightly synchronized with their cognitive progress.

5. **Evaluation in Free Recall** - Within the study spaces, the Free Recall tab lets students type out everything they can remember about a lesson without looking at their notes. To grade this automatically, the system uses a method called Cosine Similarity which is calculated natively inside the database using the Supabase pgvector extension. This database-level calculation removes the need for complex mathematical scripts in the

application's backend code by computing vector geometry directly on the cloud server.

When a student uploads their study materials, the application breaks the document into small pieces and converts those pieces into lists of numbers called high-dimensional vector embeddings that capture their exact meaning. When the student types out their recall summary, the system converts their typed text into a vector too.

Instead of running slow calculations on the backend application layer, the backend simply sends the student's vector to Supabase. The database natively calculates the mathematical angle between the student's typed vector $\vec{A}$ and the lesson's stored vector $\vec{B}$ using the Cosine Distance Operator <=>, where the final Cosine Similarity is found through the simple subtraction formula:

$$Cosine\ Similarity\ =\ 1\ -\ (A\ <=>\ B)$$

If the database returns a score close to 1, it means the student's written ideas point in the same directional meaning as the lesson. If the similarity score falls below a set quality threshold (such as 0.75), the system identifies that a gap exists. The system then compares the student's typed text directly against the lesson's key glossary terms to spot exactly which keywords were missing, displaying this feedback immediately to the student so they know what to review.

6. **Feedback generation in Feynman Technique** - The Feynman Technique helps students learn by explaining hard topics in simple, everyday language, as if they are teaching a five-year-old child. To evaluate how well a student does this, the application guides a large language model (LLM) through a three-step evaluation pipeline.

   The first step is the Jargon Check. The AI scans the student's written explanation and compares it against the main glossary list extracted during the document upload phase. If the student uses highly complex terms (like "photosynthesis" or "electromagnetism") without breaking them down simply, the AI flags those words and tells the student they need to explain them in simpler terms.

   The second step is Analogy Verification. The AI checks if the student used a helpful comparison or everyday story (an analogy) to make the idea clear. For example, if explaining electrical circuits, the AI checks if the student compared it to water flowing through pipes to make the concept easier to grasp.

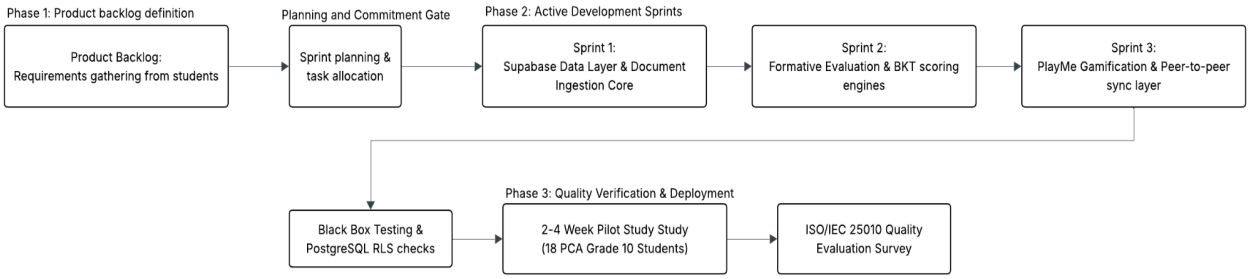
   The third and final step is Finding the Gaps. The AI compares the student's simple explanation to the original study text to make sure they did not make a factual error or skip an important step. If the student made a mistake, the AI does not just give them a low grade. Instead, it acts like a curious child and asks simple, guiding questions (such as, "But how

does the water get up into the sky in the first place?") to help the student find and correct their own learning gaps.

SE PARADIGM

This study utilizes the Agile Scrum framework to manage the iterative design, implementation, and deployment of LearnSync AI. Because the platform combines complex and highly interactive features such as real-time Bayesian Knowledge Tracing (BKT) metrics updates, large language model (LLM) semantic evaluations, procedural game generation, and peer-to-peer database cloning, a rigid, sequential development model is highly impractical. Empirical software research proves that traditional linear models struggle to accommodate the unpredictable parameter adjustments and runtime optimizations required by artificial intelligence microservices and database-level security policies (Jali et al., 2026; Daraojimba et al., 2024). By adopting Agile Scrum, the development team can construct the platform in short, manageable sprint iterations, allowing for immediate validation of the machine learning algorithms and secure database rules within a tight schedule (Akgun & Greenhow, 2021; Jing et al., 2023). This adaptive configuration is essential for an educational companion like LearnSync AI, where frontend user telemetry must connect flawlessly with active database schemas without latency or data loss.

Figure 2. *Agile SDLC*



The execution of the Agile Scrum framework across the development lifecycle is structured into the following sequential phases:

1. **Phase 1: Product Backlog** - During this initial phase, the research team establishes the master list of software requirements from the perspective of the system's single user role: the student. These requirements are formalized as specific functional specifications to keep the development process aligned with the needs of the students. The students must be able to create secure, private study spaces where they can upload text-only reference PDFs (maximum 20 MB) to act as custom knowledge repositories. Additionally, students should have access to four self-paced study options within their space container, including adaptive flashcards, free recall summaries with immediate semantic coverage feedback, Feynman Technique explanations, and an AI-driven course resource chatbot. For structured evaluations, students need the ability to generate

customized, adaptive practice tests of various formats (multiple-choice, matching, fill-in-the-blank, identification, and true/false) that scale in difficulty based on their real-time BKT mastery state. Finally, the student backlog mandates a gamified study review engine containing crosswords, matching cards, and timed trivia that dynamically adjust in difficulty using the student's BKT scores, alongside a secure peer-to-peer sharing utility where students can share their spaces for read-only peer viewing.

2. **Planning and Commitment Stage** - Before committing to code generation, the development team transition requirements through a strict Planning and Commitment Gate. In this collaborative phase, the team conducts sprint planning sessions to break down the product backlog specifications into modular, highly actionable engineering tasks. This process includes estimating developmental complexities, designating feature priorities, allocating tasks among programmers, and establishing a strict sprint calendar. By establishing clear commitments at this stage, the team ensures that the development cycles remain tightly focused, preventing scope creep and ensuring that high-risk mathematical models and data security layers are prioritized before active coding begins.

3. **Phase 2: Active Development Sprints -** The second phase represents the core coding lifecycle of the application, structured into three distinct, iterative sprints:

    a. **Sprint 1: Supabase Data Layer & Document Ingestion Core**: This sprint focuses entirely on establishing the core cloud infrastructure, database schemas, and text-parsing pipelines needed to handle student document uploads. The development team configures the PostgreSQL tables on Supabase to house the structural components of the data stores, including the parent study spaces tables and the child learning materials and embeddings records. Strict PostgreSQL Row-Level Security (RLS) policies are enforced to guarantee that each student's profile data, study materials, and metrics remain completely private and isolated under their unique authenticated user ID. Finally, the team implements text extraction scripts using large language models to read uploaded PDFs, extract concepts & definitions, and generate semantic vectors, saving the parsed results directly into the database.

    b. **Sprint 2: Formative Evaluation & BKT Scoring Engines**: This sprint builds the active learning algorithms, the predictive cognitive modeling engine, and the Generative AI evaluation services. The software team programs the

classical four-parameter Bayesian Knowledge Tracing (BKT) scoring calculations to dynamically track the latent knowledge state of the student after each question attempt (0 for incorrect, 1 for correct), updating the mastery profiles in the tracking database. The team develops the routing algorithm that queries these baseline metrics and dynamically adjusts the difficulty of generated test questions to match the student's current conceptual limits. The development phase also establishes the semantic evaluation queries utilizing vector databases to calculate similarity thresholds for Free Recall text inputs, integrates the language model prompt templates to evaluate Feynman simple-terms explanations and chatbot queries, and logs the activity transaction timestamps directly into the unified Activity and Streak Logs database to track daily engagement.

c. **Sprint 3: PlayMe Gamification & Peer-to-Peer Sync Layer**: This sprint focuses on the motivational design, peer-to-peer workspace sharing, and the user interface layout. The developer team codes the backtracking layout algorithm that dynamically structures crossword grids, trivia questions, and vocabulary matches based on the parsed glossaries, using the student's BKT score as a difficulty

reference. The gamification and streak engine evaluates a unified database log of both active study space reviews and game sessions to validate milestone badge criteria (such as a badge for reaching a 10-day streak) and check daily study log timestamps to manage and update active consecutive daily streaks. The collaboration logic is programmed to generate secure synchronization tokens and configure read-only database permission layers to allow classmates to safely clone spaces. The sprint concludes by compiling the backend API gateways and real-time database channels with a responsive, single-role web frontend student interface.

4. **Phase 3: Quality Verification & Development** - During this phase, the completed LearnSync AI web application is subjected to rigorous testing and validation procedures to ensure absolute stability and performance before classroom rollout. The phase initiates with extensive Black Box Testing and PostgreSQL Row-Level Security checks, designed to verify that the core system modules operate with complete technical accuracy. These procedures confirm that PostgreSQL database permission layers successfully isolate data, preventing student accounts from reading, editing, or deleting unauthorized classmates' spaces. Following structural security and algorithmic validation, the web

application is formally launched for a live, two-to-four-week pilot study of 18 Grade 10 students at Pasig Christian Academy. These students independently use the web platform to study and review their actual secondary education curriculum materials outside the classroom, providing a realistic self-study environment for tracking active usage patterns and system performance. At the conclusion of this active pilot window, a structured software quality survey is administered to the participating students. This evaluation tool translates user experience feedback into objective, numerical averages using a 5-point Likert scale to formally measure and verify the application's overall operational quality across four core software criteria, specifically evaluating its Functional Suitability, Usability, Performance Efficiency, and Security under the standardized ISO/IEC 25010 framework.

DATA FLOW DIAGRAM

This section defines the LearnSync AI's Data Flow Diagrams (DFDs). These visual models map the recursive flow of course materials, study responses, and gamified data, charting how information moves seamlessly from raw ingestion to the generation of adaptive assessment questions and real-time learning analytics

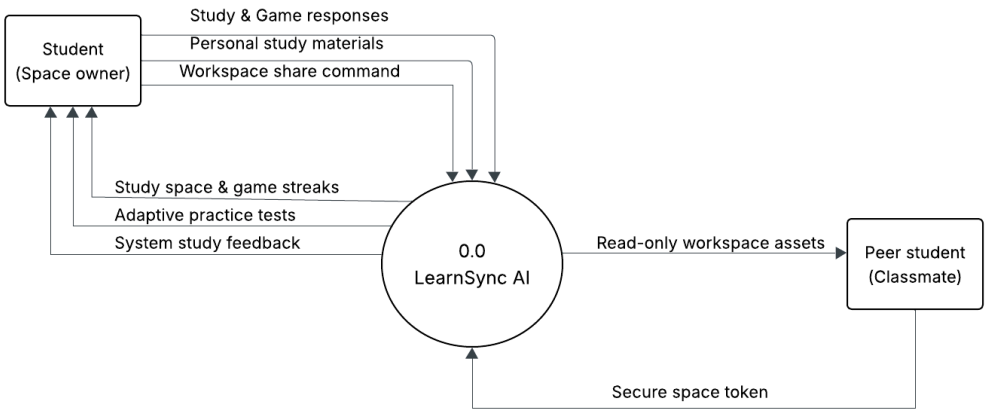
1. **Level 0: Context Diagram -** When using the system, the Student (Space owner) sends three main types of inputs into the LearnSync AI platform. First, they provide Study & Game responses, which are the active clicks and typing they do while reviewing. This includes rating card retrieval difficulty (by choosing Easy, Medium, or Hard) during flashcard reviews, typing out summaries for recall exercises, and interacting with study games. Second, they upload their Personal study materials, which are PDF files up to 20 MB containing the student's lessons, textbooks, or notes. Lastly, they send the Workspace share command, which is just the user clicking a button in the app to make their private workspace public so a classmate can view it.

   In return, the platform delivers three main types of outputs back to the Student (Space owner). The Study space & game streaks are live screen updates that show their total study hours, daily streaks, and digital badges. The Adaptive practice tests are customized quizzes that automatically get easier or harder depending on how well the student understands the topic. Finally, the System study feedback provides simple, helpful advice on their progress, showing them what key terms they missed during active recall or confirming if their simplified explanations make logical sense.

   The Peer student (Classmate) interacts with the system using two simple data flows. To view a shared workspace, they must first send a Secure space token, which is a unique link or access code that acts as a

secure passkey. Once the system checks and verifies this code, it displays the Read-only workspace assets on their screen. This data flow allows the classmate to read and study the shared flashcards, summaries, and documents. Crucially, this access is strictly view-only, meaning the peer student can read the materials but cannot edit them or copy the setup into their own account.

Figure 3. *Level 0 Context Diagram*



2. **Level 1: Date flow diagram -** the system's internal work is divided into five separate sub-processes, three database storage areas (called data stores), and two groups of users: the Student (Space owner) and the Peer Student (Classmate). By splitting the system this way, the diagram shows how uploading files, reading text with AI, and tracking learning progress all work together securely.

The first sub-process, 1.0 Space Access and Space sharing Control, handles the privacy settings of study spaces and controls how students share their work with classmates. When a Student (Space owner) wants to share a study space, they trigger a Workspace share command. This tells the process to create a secure key and run a Space configuration update & token query to save that key in the Study spaces, Learning materials and Embeddings data store. When a Peer Student (Classmate) wants to view this shared space, they enter a Secure Space token. The process sends this token to the database, which replies with a Space sharing status response to confirm the key is correct. Once verified, Process 1.0 displays the Read-only study space on the classmate's screen. This ensures classmates can study the shared materials but cannot edit, delete, or change any of the original student's files.

The second sub-process, 2.0 Document Parsing and Semantic Ingestion, manages how new study files are uploaded and prepared. When a Student (Space owner) uploads their Personal Study Materials (as a PDF up to 20 MB) inside a study space, this process reads the document. It finds the main keywords, creates a glossary list, and converts the sentences into a math format that AI can read. It then saves these results directly to the Study spaces, Learning materials and Embeddings data store using the Parsed text, keywords and vectors data flow. The system automatically links every uploaded file and keyword list to the main

study space's ID, making sure the materials are immediately organized and ready for the student to practice with.

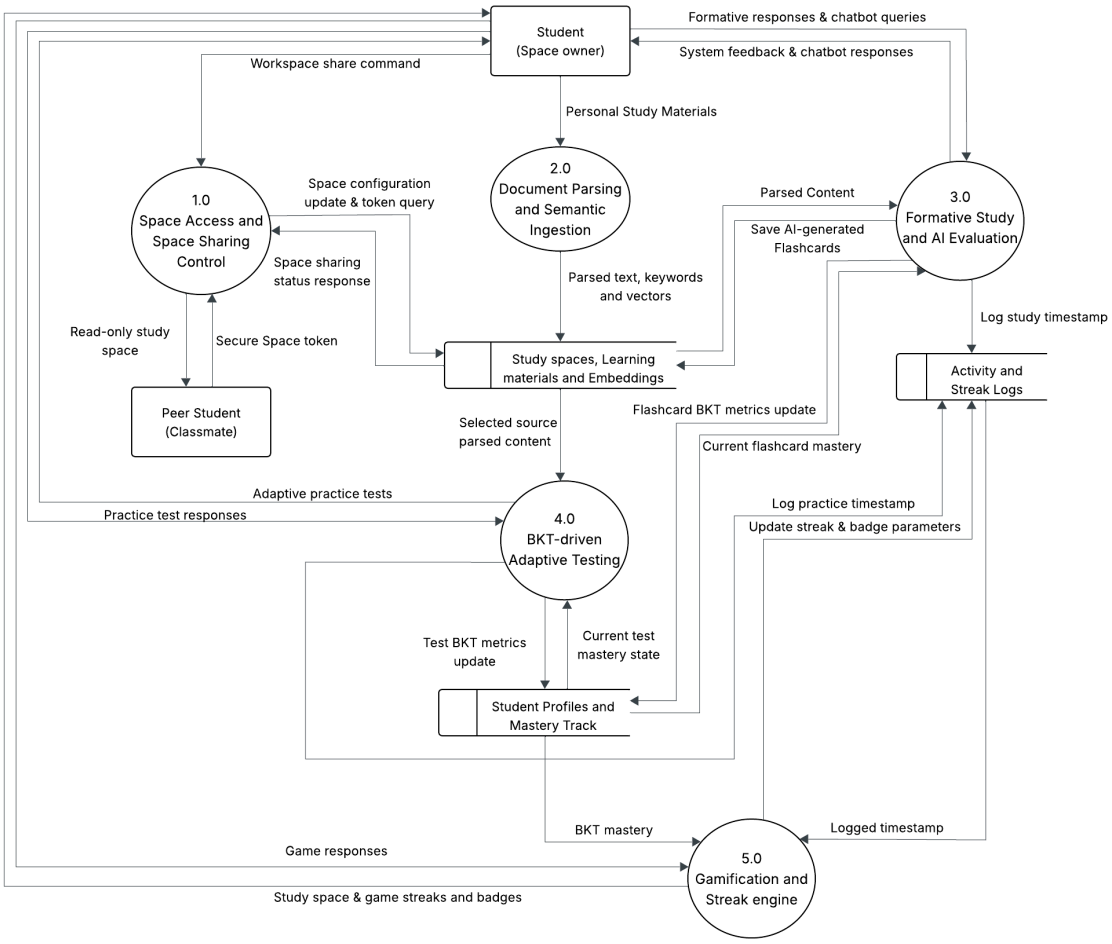
The third sub-process, 3.0 Formative Study and AI Evaluation, runs the non-graded review activities like the Feynman Technique, Free Recall, flashcard generation, and chatbot interactions. When the Student (Space owner) inputs their Formative responses & chatbot queries (such as asking questions to the AI, summarizing topics, or rating flipped flashcard difficulty as Easy, Medium, or Hard), the process pulls the original Parsed Content from the Study spaces, Learning materials and Embeddings data store. Using AI, the process calculates how much of the topic the student successfully remembered, provides immediate answers, and sends back System feedback & chatbot responses to the user's interface. To handle interactive flashcards, this process reads the student's history using the Current flashcard mastery flow from the Student Profiles and Mastery Track database. It converts the student's qualitative ratings (Easy and Medium as a correct recall, and Hard as an incorrect recall) into standard binary observations ($obs_t = 1$) or ($obs_t = 0$). It also routes the correct rating classification (such as mapping Easy to a low guess rate of 0.05 and Medium to a moderate guess rate of 0.25) into the update pipeline, updates it using the Flashcard BKT metrics update flow, and saves the updated states back to the database. When the student finishes reviewing, the process sends a Log study timestamp to the Activity and Streak Logs data store to record that they studied.

The fourth sub-process, 4.0 BKT-driven Adaptive Testing, serves as the main center for grading practice tests and choosing the next questions. When the Student (Space owner) answers multiple-choice questions or flashcards, their answers are captured as Practice test responses (marked as a 1 for correct or a 0 for incorrect). The process retrieves the original study files using the Selected source parsed content flow from the Study spaces, Learning materials and Embeddings data store, and reads the student's learning history using the Current test mastery state flow from the Student Profiles and Mastery Track data store. Using the Bayesian Knowledge Tracing (BKT) math model, the process calculates what the student knows and automatically sends back Adaptive practice tests with harder or easier questions. At the same time, it updates the student's profile in the database using the Test BKT metrics update flow and saves the test time in the Activity and Streak Logs database using a Log practice timestamp.

The fifth and final sub-process, 5.0 Gamification and Streak engine, keeps students motivated by awarding badges and study streaks without affecting their actual test scores. When the Student (Space owner) plays study games, their actions are sent as Game responses. To make sure the games match the student's skills, the process reads their learning level from the Student Profiles and Mastery Track database using the BKT mastery flow as a reference to adjust game difficulty. It also reads study times from the Activity and Streak Logs database using the Logged

timestamp flow to check if the student has studied within a daily calendar window. The engine then updates the student's streak records using the Update streak & badge parameters flow and displays their updated Study space & game streaks and badges (such as active days, and earned badges) right on their dashboard screen.

Figure 4. *Level 1 Main Process*



TOOL TO TEST FOR ACCURACY

This study uses a specialized testing framework to verify that LearnSync AI's learning and grading modules work accurately, securely, and reliably. Because the platform relies on real-time artificial intelligence and probability calculations rather than simple, fixed question sets, traditional software testing is not enough. The research team must isolate both the math engine and the AI generators to verify that the system tracks student progress correctly, routes

questions appropriately, and keeps user data secure before the application is deployed in a live classroom.

To measure how well the application tracks student learning progress, the backend math engine undergoes rigorous offline testing. This step simulates student interactions using pre-recorded study histories to check if the algorithm can accurately predict whether a student has mastered a topic or is still struggling.

1. **Confusion Matrix for Students Mastery** - The system predicts a student's learning state using a simple "Mastered" versus "Not Mastered" classification. To evaluate how accurate these predictions are, the research team uses a confusion matrix. This matrix acts as a diagnostic report card, comparing the system's prediction against how the student actually performs on subsequent practice questions. The results are grouped into four clear outcomes:

    - **True Positives** (TP) : The system predicts that the student has mastered a concept, and the student proves this correct by answering the next question correctly.

    - **True Negatives** (TN) : The system predicts that the student has not yet mastered a concept, and the student confirms this by answering the next question incorrectly (pointing out an active learning gap).

- **False Positives** (FP) : The system incorrectly predicts that the student has mastered a concept, but the student gets the next question wrong because the material is actually too difficult.
- **False Negatives** (FN) : The system incorrectly predicts that the student is struggling, but the student gets the next question right because they actually understand the concept.

This confusion matrix is chosen because evaluating the system's decisions using a basic average percentage is not enough. A simple average hides the exact nature of the system's errors. By separating these four outcomes, the research team can verify that the system's mastery boundaries are highly optimized.

From these outcomes, the development team calculates standard testing metrics. Precision measures the system's reliability, ensuring that when the application claims a student has mastered a topic, that claim is highly trustworthy (minimizing false alarms that push students into frustratingly difficult questions):

$$Precision = \frac{TP}{TP + FP}$$

Conversely, Recall measures the system's sensitivity, ensuring that the application successfully catches every student who is genuinely struggling so that no active learning gaps are ignored (minimizing missed opportunities that trap students in boring, repetitive lessons):

$$Recall = \frac{TP}{TP + FN}$$

To measure the overall balance between reliability and sensitivity, the team calculates the harmonic mean of precision and recall, yielding the F1-Score:

$$F_1 = 2\,x\,\frac{Precision\,x\,Recall}{Precision + Recall}$$

2. **Cognitive Tracking Accuracy via Mean Absolute Error** (MAE) - While the confusion matrix checks high-level, yes-or-no mastery decisions, Mean Absolute Error (MAE) measures the accuracy of the shifting percentages calculated inside the Bayesian Knowledge Tracing (BKT) engine. Learning is not a simple, linear process; a student's level of understanding rises and falls continuously. MAE acts like a statistical tape measure, calculating the average mathematical distance between the system's predicted probability of a student answering correctly and the student's actual, real-world outcome ($obs_t \in \{0, 1\}$)

   MAE is selected as a companion metric to the confusion matrix because it evaluates the fine-grained math updates happening inside the algorithm. It calculates how close the system's percentage guesses are to real-world outcomes. Averaging these absolute errors across the entire dataset provides a transparent, unweighted measurement of prediction error.

The MAE across a series of *M* total student practice attempts is calculated using the following formula:

$$MAE = \frac{\Sigma P(osb_t = 1) - obs_t}{M}$$

Where:

- *M* represents the total number of sequential practice attempts evaluated across the study sessions.
- $P(obs_t = 1)$ represents the algorithm's predicted probability that the student will answer correctly at learning opportunity t, calculated as:

$$P(obs_t = 1) = P(L_t)(1 - P(S)) + (1 - P(L_t))P(G)$$

- obs_t represents the actual, real-world outcome of the student's attempt, where a correct response is recorded as 1 and an incorrect response is recorded as 0.

By keeping the baseline MAE as close to zero as possible during system testing, the engineering team prevents the math engine from drifting and ensures that the app's internal cognitive tracker closely mirrors the actual learning speeds of real students.

3. **Black-Box Functional and Security Testing** - In addition to validating the mathematical models, the development team conducts black-box testing. This method evaluates the system's inputs and outputs without needing to inspect the underlying source code. Semantic parser testing verifies that when a student uploads a standard, text-only PDF study module, the system successfully extracts a clean, organized list of key terms and questions matching the database layout without any broken characters. In parallel, database query validation checks the system's search operators to ensure that semantic vector searches accurately match user search queries with relevant study content during Free Recall activities. To verify the adaptive intelligence layer, routing logic testing confirms that a sequence of incorrect answers successfully lowers the student's BKT score and prompts the AI generator to route the student to simpler, foundational-tier questions. To ensure data safety, peer-to-peer sharing permission validation checks that security rules on database storage buckets function perfectly, confirming that when a student uses a shared token to view a classmate's study space, they are strictly locked out from editing, deleting, or altering any materials. Finally, role-based access and isolation testing verifies that database policies completely isolate individual user accounts, ensuring that a logged-in student can only view, create, or update data tied to their unique, verified user ID, making it impossible for classmates to access another student's private study metrics.

4. **ISO/IEC 25010 Quality Model Survey** - The technical testing phase concludes with a software quality evaluation. Grounded in the standardized ISO/IEC 25010 Software Quality Model, this descriptive evaluation is administered to the pilot cohort of 18 Grade 10 students at Pasig Christian Academy after they use the application. The evaluation gathers student feedback to measure the platform's production-readiness across four key areas, starting with functional suitability to measure how completely and accurately the system's features, such as adaptive tests, memory recall feedback, and peer-to-peer resource sharing, meet the students' actual self-study needs. The second area, usability, evaluates user interface clarity, ease of navigation, and whether the gamification makes learning engaging without causing unnecessary mental strain. Additionally, performance efficiency measures the speed of the application, tracking database update times and the responsiveness of the AI during feedback generation. The evaluation concludes with security, which formally verifies student privacy, the absolute isolation of user profiles, and the integrity of read-only permissions during peer-to-peer resource sharing sessions.

POPULATION AND SAMPLING

The target population for this study consists of grade 10 students within Pasig Christian Academy who often use traditional, standardized study materials and conventional classroom evaluation frameworks. The study focuses specifically on a secondary school academic environment transitioning into digital learning systems, where students can benefit from automated, AI-powered parsing tools and personalized adaptive pathways to improve concept retention and independent study habits.

To select the participants, a purposive sampling strategy was used to specifically utilize total enumeration of the target. This strategy isolates a specialized cohort of young learners who are navigating concept-heavy junior high school curriculum that mandate consistent independent review and cooperative academic peer validation. The active sample size comprises a single operational cohort of eighteen (18) Grade 10 students from Pasig Christian Academy. This cohort will serve as the primary end-users responsible for executing self-paced, independent review sessions using the platform's formative modules, including AI-driven Flashcards, Free Recall semantic evaluations, and Bayesian Knowledge Tracing (BKT) Adaptive Tests.

EVALUATION INSTRUMENT

The primary evaluation instrument is a researcher-designed, criteria-based system acceptability questionnaire structured to quantitatively

measure the technical execution and user satisfaction of LearnSync AI. The development of the instrument is closely patterned after standard software product quality metrics, specifically focusing on dimensions that directly impact end-user experience: functional suitability, usability, performance efficiency, and data security.

The instrument is divided into two distinct components to capture both quantitative benchmarks and qualitative insights from the 18 Grade 10 respondents:

- **Quantitative Matrix:** This section contains close-ended indicators distributed across the core technical domains. Each indicator maps to the system's operational features—such as the processing efficiency of the AI flashcard parsing engine and the layout stability of the side-by-side study spaces. To evaluate these indicators, respondents rate each item using a structured, forced-choice 4-point rating scale. This scale eliminates a neutral midpoint to ensure definitive data, utilizing the following criteria:

    - **4 – Strongly Agree (SA) / Excellent:** The system attribute functions perfectly, providing exceptional utility, high security, and seamless navigation.
    - **3 – Agree (A) / Good:** The system attribute meets all technical requirements and operates efficiently with minor or negligible usability friction.

    - **2 – Disagree (D) / Fair:** The system attribute meets minimum requirements but exhibits notable performance lags or unintuitive interface layouts.
    - **1 – Strongly Disagree (SD) / Poor:** The system attribute fails to meet baseline operational standards, presenting critical errors or complete layout failures.

- **Qualitative Feedback Loop:** Positioned at the end of the instrument is an open-ended feedback section. This component allows student respondents to explicitly detail unexpected software bugs, system latency issues, and user interface preferences that are not fully captured by numerical ratings.

DATA COLLECTION METHOD

To systematically capture performance metrics and user feedback for LearnSync AI, data collection will be executed through a controlled, three-phase framework. All research instruments (including the pre-test, post-test, and the system acceptability questionnaire) will be deployed using Google Forms and administered personally by the researchers to ensure optimal data integrity. The physical, in-person distribution strategy allows the researchers to directly coordinate the testing environment, guide the 18 Grade 10 respondents through

the application's interface, and immediately address any technical or procedural questions in real time.

The researchers will personally meet with the 18 Grade 10 respondents in a designated school setting at Pasig Christian Academy. After explaining the purpose of the study and securing necessary academic permissions, the researchers will provide a Google Form link containing the diagnostic pre-test. This test will be answered in-person to establish a controlled baseline measurement of the students' current conceptual retention and topic mastery before interacting with the software.

Following the baseline assessment, the researchers will personally introduce the LearnSync AI platform to the cohort. The 18 student respondents will be given access to the application to engage with its core operational modules, including the AI-driven Flashcards, Free Recall spaces, and BKT Adaptive Tests.

Upon conclusion of the system pilot period, the final data collection phase will be conducted in person. The researchers will distribute the remaining Google Forms to the respondents to collect final comparative data. The Evaluative Post-Test will be administered to measure the quantitative shift in academic retention following platform usage. After the Post-test, the evaluation survey would be given to the students. The 4-point rating scale and qualitative feedback questionnaire will be answered by the students to record their direct evaluation of

the system's functional suitability, usability, performance efficiency, and data security.

By utilizing Google Forms distributed personally, the researchers combine the automated logging and data organization benefits of a digital survey tool with the high response rates and strict environment control of face-to-face administration.

ETHICAL CONSIDERATIONS

To safeguard student participants and uphold academic integrity, the implementation of LearnSync AI strictly adheres to the Data Privacy Act of 2012 (RA 10173). The platform enforces the principle of data minimization by solely collecting the basic academic metadata, strictly necessary for calculating Bayesian Knowledge Tracing (BKT) metrics and generating personalized content. To ensure absolute privacy, the cloud infrastructure leverages Supabase to enforce strict PostgreSQL Row-Level Security (RLS) policies, which completely isolates user profiles and ensures a logged-in student can only view or modify data tied directly to their unique authenticated ID.

Furthermore, while the system enables a peer-to-peer collaboration layer, data integrity is tightly maintained through secure, system-generated space tokens that grant classmates strictly read-only permissions. This structural safeguard ensures that authorized peers can view shared flashcards and

summaries but are strictly locked out from editing, deleting, or altering the original student's private workspace assets and tracking metrics. Finally, all collected information remains strictly confidential, meaning student data is securely stored within isolated repositories and is never shared, leased, or sold to third-party entities.

SUMMARY

This chapter details the methodology, technical architecture, and validation framework used to design, implement, and evaluate the LearnSync AI web platform. The study employs a developmental research design utilizing a quantitative approach to track system creation across three distinct phases: analysis and design, active technical development, and live pilot evaluation. The system operates on a student-centric, three-tier model designed to isolate workloads across clear processing layers. The client-facing presentation layer runs natively within the browser to manage user interaction points like the Study Space Hub, PlayMe Portal, and the main analytics dashboard. This routes directly to the application layer, which acts as the computational brain executing the formative parsing engine, the Bayesian Knowledge Tracing scoring engine, and the procedural gamification modules. Persistent data operations are managed in the data tier powered by a unified Supabase platform, deploying a PostgreSQL database with strict Row-Level Security, a pgvector extension for

semantic search, and secure file-bucket storage for uploaded PDF learning materials.

To deliver a highly personalized experience, the system implements the classical four-parameter Bayesian Knowledge Tracing algorithm. By calculating the mathematical variables for prior knowledge $P(L_0)$, learning rate $P(T)$, guess probability $P(G)$, and slip probability $P(S)$, the engine converts binary student responses into live mastery updates that dynamically adjust spaced repetition intervals, assessment difficulties, and gamification configurations. The development team utilizes the Agile Scrum framework to manage these iterative artificial intelligence microservices and database configurations across distinct sprints covering core ingestion infrastructure, evaluation engines, and a secure peer-to-peer cloning layer. Modeled workflows map these data transactions between system processes and users, tracking user inputs to generate adaptive practice tests, streak telemetry, and formative system feedback, while simultaneously handling a secure passkey routing layer for read-only peer asset validation.

Technical evaluation isolates the predictive modeling engines using a confusion matrix to track precision, recall, and F1-score, alongside Mean Absolute Error calculations to ensure mathematical updates closely mirror real student learning speeds. System acceptability is further quantified using an ISO/IEC 25010 software quality model survey. Using a purposive sampling strategy with total enumeration, the pilot study monitors a single operational cohort consisting exactly of an eighteen student sample size of Grade 10

students from Pasig Christian Academy. Data collection uses a structured, personal three-phase distribution of Google Forms featuring an in-person diagnostic pre-test, live pilot access for the 18 student participants, and a concluding evaluative post-test alongside a 4-point Likert scale survey. Finally, in strict compliance with the Data Privacy Act of 2012 (RA 10173), the platform enforces data minimization, isolates student profiles via database security policies, and hard-locks shared peer workspaces to read-only permissions via system tokens to prevent unauthorized data mutation.

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